Classwork – Descriptive statistics

You do not need to submit Warmup exercise 1A – 1D in Moodle.

This exercise introduces common ways of presenting frequency data.

$$f_i = \text{number of observations in category/value } i, \qquad p_i = \frac{f_i}{n},$$

where n is the total number of observations. Check that

$$\sum_i f_i = n, \qquad \sum_i p_i = 1.$$

Cumulative frequency and cumulative relative frequency are

$$F_k = \sum_{i=1}^{k} f_i, \qquad P_k = \sum_{i=1}^{k} p_i.$$

Warmup 1A – Line graph and bar graph

Illustrative starting salaries of recently graduated Finnish engineers (EUR/month):

```
salary = [3600,3700,3800,3900,4000,4100,4200,4300,4500,4700,5000]
frequency = [2,3,5,6,8,5,4,3,2,1,1]
```

Create a line graph and a bar graph. Verify the total number of observations.

Python code for this example. Save as `Ex1A_line_bar.py`.

```
import matplotlib.pyplot as plt
salary = [
    3600, 3700, 3800, 3900, 4000,
    4100, 4200, 4300, 4500, 4700, 5000
]

frequency = [
    2, 3, 5, 6, 8,
    5, 4, 3, 2, 1, 1
]

print("\nEXERCISE 1 - PART A")
print("Total number of observations =", sum(frequency))

# Line graph
plt.figure(figsize=(9, 5))
plt.plot(salary, frequency, marker="o")
plt.xlabel("Starting salary (EUR/month)")
plt.ylabel("Frequency")
plt.title("Starting Salaries of Recently Graduated Engineers")
plt.grid(True)
plt.show()

# Bar graph
plt.figure(figsize=(9, 5))
plt.bar(salary, frequency, width=70)
plt.xlabel("Starting salary (EUR/month)")
plt.ylabel("Frequency")
plt.title("Starting Salaries of Recently Graduated Engineers")
plt.show()
```

Warmup 1B – Pie chart: relative frequencies

A survey was conducted among 100 recently graduated engineers. They were asked which type of employer they currently work for.

The responses were:

Employer type	Frequency
Private company	30
Public sector	25
University/research	20
Self-employed	25

Let f_i denote the frequency of category i, and let n denote the total number of observations.

For each category:

1. Calculate the relative frequency
$$p_i = \frac{f_i}{n}.$$
2. Express the relative frequency as a percentage,
$$100p_i\%.$$
3. Check that the relative frequencies sum to 1 (or that the percentages sum to 100%).

Then create a pie chart showing the relative frequencies.

Save your Python program as `Ex1B_pie.py`.

```
import matplotlib.pyplot as plt

categories = [
    "Private company",
    "Public sector",
    "University/research",
    "Self-employed"
]

frequency = [30, 25, 20, 25]

print("Total number of observations =", sum(frequency))

plt.figure()
plt.pie(
    frequency,
    labels=categories,
    autopct="%1.1f%%"
)
plt.title("Employer Types of Recently Graduated Engineers")
plt.show()
```

Warmup 1C – Histogram

The following data show the commuting times, in minutes, of 10 recently graduated engineers travelling from home to work:

$$10,\ 12,\ 15,\ 18,\ 20,\ 21,\ 25,\ 27,\ 30,\ 32.$$

Here, each value is one observation of the numerical variable *commuting time*.

A histogram groups numerical observations into intervals (*bins*) and shows how many observations fall into each interval.

1. Determine the number of observations n.
2. Find the minimum and maximum commuting times.
3. Create a histogram using 5 bins.
4. Use the histogram to describe how the observations are distributed over the range of commuting times.

Save your Python program as `Ex1C_histogram.py`.

```
import matplotlib.pyplot as plt

commuting_time = [
    10, 12, 15, 18, 20,
    21, 25, 27, 30, 32
]

print("Number of observations =", len(commuting_time))
print("Minimum =", min(commuting_time))
print("Maximum =", max(commuting_time))

plt.figure()
plt.hist(
    commuting_time,
    bins=5,
    edgecolor="black"
)
plt.xlabel("Commuting time (minutes)")
plt.ylabel("Frequency")
plt.title("Commuting Times of Recently Graduated Engineers")
plt.show()
```

Warmup 1D – Cumulative frequency

Use the same commuting-time data as in Part C:

$$10,\ 12,\ 15,\ 18,\ 20,\ 21,\ 25,\ 27,\ 30,\ 32.$$

Instead of showing the frequency in each interval separately, we now consider the *cumulative frequency*.

If f_i is the frequency in interval i, the cumulative frequency up to interval k is

$$F_k = \sum_{i=1}^{k} f_i.$$

Thus, F_k tells us how many observations have accumulated up to that interval.

The final cumulative frequency must equal the total number of observations:

$$F_{\text{final}} = n.$$

1. Create a cumulative histogram using the same 5 bins as in Part C.
2. Check that the final cumulative frequency equals n.
3. Use the graph to describe how the observations accumulate as commuting time increases.

Save your Python program as `Ex1D_cumulative.py`.

```
import matplotlib.pyplot as plt

commuting_time = [
    10, 12, 15, 18, 20,
    21, 25, 27, 30, 32
]

n = len(commuting_time)

print("Number of observations =", n)

plt.figure()
plt.hist(
    commuting_time,
    bins=5,
    cumulative=True,
    edgecolor="black"
)
plt.xlabel("Commuting time (minutes)")
plt.ylabel("Cumulative frequency")
plt.title("Cumulative Histogram of Commuting Times")
plt.show()
```

2A – Gasoline prices in San Francisco

Let f_i be the frequency of value i. The relative frequency is

$$p_i = \frac{f_i}{n}, \qquad \sum_i p_i = 1.$$

The sample is

$$3.88, 3.90, 3.93, 3.90, 3.93, 3.96, 3.88, 3.94, 3.96, 3.88, 3.94, 3.99, 3.98.$$

Represent it as (a) a frequency table and (b) a relative-frequency line graph.

2B – Gasoline prices in the Helsinki region

Find 14 observations of 95 E10 gasoline prices in the Helsinki region from August 2026, in EUR/litre, from https://www.hintatutka.net/suomi/uusimaa/helsinki. Use the same fuel type throughout and record the date and station for every observation.

1. Enter the 14 prices into a Python list.
2. Construct a frequency table.
3. Calculate relative frequencies.
4. Check mathematically that $\sum_i f_i = 1$.
5. Draw a relative-frequency line graph.
6. Compare with the San Francisco example.
7. Ask whether the result makes sense.

Important. You must also understand the mathematical formulas. (E.g., in a good exercise submission, some calculations will be done with paper and pencil). AI and programs can make mistakes; an engineer must be able to recognize results that do not make sense.

3 – Road Traffic Accidents in Finland

Source: Statistics Finland, PxWeb, Table 12sc, years 2003–2025. The 2025 data are preliminary.
https://pxdata.stat.fi/PxWeb/pxweb/en/StatFin/StatFin__ton/12sc.px/

Data

```
year = [2003,2004,2005,2006,2007,2008,2009,2010,2011,2012,2013,2014,
2015,2016,2017,2018,2019,2020,2021,2022,2023,2024,2025]
accidents = [6907,6767,7022,6740,6657,6881,6414,6072,6408,5725,5334,
5324,5185,4752,4432,4312,4002,3608,3243,3110,2944,2811,2638]
fatalities = [379,375,379,336,380,344,279,272,292,255,258,229,270,258,
238,239,211,223,225,196,185,181,193]
```

Formulas required

For observations $x_1, \ldots, x_n$,

$$
\begin{aligned}
f_i &= \text{number of observations having value } x_i, \\
p_i &= \frac{f_i}{n}, \qquad\qquad \sum_i p_i = 1, \\
P_k &= \sum_{i=1}^{k} p_i, \\
\bar{x} &= \frac{1}{n}\sum_{i=1}^{n} x_i, \\
s &= \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}.
\end{aligned}
$$

For $x_{(1)} \le \cdots \le x_{(n)}$,

$$
\text{median} = \begin{cases} x_{((n+1)/2)}, & n \text{ odd}, \\ \dfrac{x_{(n/2)} + x_{(n/2+1)}}{2}, & n \text{ even}. \end{cases}
$$

The mode is the most frequently occurring value. Use $n-1$ for the sample standard deviation.

3A – Yearly road traffic accidents

The variable `accidents` gives the number of road traffic accidents in Finland for each year from 2003 to 2025.

(a) Create a line graph with year on the horizontal axis and the number of accidents on the vertical axis.

(b) Find the sample mean, median, minimum, maximum, and sample standard deviation of the annual accident counts.

(c) Describe the development of road traffic accidents over time. Do not only report the calculated statistics. What does the graph show?

3B – Road traffic fatalities

The variable `fatalities` gives the number of road traffic fatalities for the same years, 2003–2025.

(a) Create a line graph with year on the horizontal axis and the number of fatalities on the vertical axis.

(b) Find the sample mean, median, minimum, maximum, and sample standard deviation of the annual fatality counts.

(c) Describe the development of road traffic fatalities over time.

3C – Comparing accidents and fatalities

The two variables refer to the same years. For example, the first value in `accidents` and the first value in `fatalities` both refer to 2003.

Compare the two time series.

(a) Plot accidents and fatalities against year.

(b) Compare the overall patterns. Do accidents and fatalities appear to increase or decrease over the period?

(c) Identify any years in which the two series appear to behave differently from their general patterns.

(d) Based on the graphs, do years with relatively high numbers of accidents also appear to have relatively high numbers of fatalities?

(e) Can the graphs alone tell you that a change in the number of accidents *causes* a change in the number of fatalities? Explain briefly.

3D Check your results

For each numerical result, ask:

- Which formula did you use?
- Does your Python code implement the same calculation?
- Does the result make sense in the context of the data?

For each graph, make sure that you know what the horizontal and vertical axes represent.

Code is not the formula.

4 – Evaporation Rate

A chemical engineer studies the evaporation of water from brine evaporation beds. The following data give the amount of evaporation, in centimetres, observed during 55 summer days spread over four years.

```
evaporation = [
    0.02, 0.06, 0.11, 0.14, 0.21, 0.21, 0.21, 0.23, 0.23, 0.24, 0.25,
    0.25, 0.25, 0.26, 0.29, 0.30, 0.30, 0.32, 0.32, 0.32, 0.33, 0.33,
    0.33, 0.33, 0.34, 0.34, 0.35, 0.35, 0.35, 0.36, 0.36, 0.37, 0.38,
    0.39, 0.40, 0.41, 0.42, 0.42, 0.42, 0.43, 0.44, 0.44, 0.44, 0.45,
    0.45, 0.45, 0.47, 0.48, 0.48, 0.48, 0.49, 0.49, 0.52, 0.55, 0.56
]
```

Formulas required in the exam

For observations $x_1, x_2, \ldots, x_n$, the sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

The sample standard deviation is

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}.$$

For ordered observations

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)},$$

the sample median is

$$\text{median} = \begin{cases} x_{((n+1)/2)}, & n \text{ odd}, \\ \dfrac{x_{(n/2)} + x_{(n/2+1)}}{2}, & n \text{ even}. \end{cases}$$

Observations within one sample standard deviation of the mean satisfy

$$\bar{x} - s \leq x_i \leq \bar{x} + s.$$

If k of the n observations are in this interval, their percentage is

$$\frac{k}{n} \cdot 100\%.$$

Tasks

(a) **Sample mean**

Find the sample mean evaporation $\bar{x}$.

(b) **Sample median**

Find the sample median evaporation.

(c) **Sample standard deviation**

Find the sample standard deviation s.

(d) **Shape of the distribution**

Create a histogram of the evaporation data.

Use the histogram to describe the shape of the distribution. Consider, for example:

- Is the distribution approximately symmetric or skewed?

- Is it approximately bell-shaped?
- Are there observations that appear unusually far from the others?
- Are the sample mean and median reasonably close to each other?

(e) **Shapiro–Wilk test for normality**

A histogram gives a visual assessment of the distribution. A statistical test can be used as an additional way of assessing whether the data are reasonably consistent with a normal distribution.

The Shapiro–Wilk test compares two hypotheses:

$$H_0 : \text{the evaporation data are consistent with a normal distribution,}$$
$$H_1 : \text{the evaporation data are not consistent with a normal distribution.}$$

Here, H_0 is the *null hypothesis*: the hypothesis being tested. The p-value tells us how unusual a test result at least this extreme would be if H_0 were true.

Use the significance level

$$\alpha = 0.05.$$

Compare the p-value with α:

$$\begin{aligned} p > 0.05 \quad &\Rightarrow \quad \text{do not reject } H_0, \\ p \leq 0.05 \quad &\Rightarrow \quad \text{reject } H_0. \end{aligned}$$

Report the p-value and state your decision.

Then interpret the Shapiro–Wilk test **together with the histogram and your comparison of the mean and median**.

Remember that not rejecting H_0 does not prove that the data follow a normal distribution. It means that the sample does not provide sufficient evidence to reject the normality assumption.

(f) **Observations within one sample standard deviation**

Use your results from parts (a) and (c) to calculate the interval

$$[\bar{x} - s,\ \bar{x} + s].$$

Count how many of the 55 evaporation observations fall within this interval.

Calculate the percentage of observations within one sample standard deviation of the mean.

Python

You may use `numpy`, `scipy.stats`, and `matplotlib.pyplot` to check your calculations and create the histogram.

For the Shapiro–Wilk test, you may use

```
from scipy.stats import shapiro

statistic, p_value = shapiro(evaporation)

print("Shapiro-Wilk statistic =", statistic)
print("p-value =", p_value)
```

Do not report only the Python output. Explain what the result means for these evaporation data.

5 – Correlation

A random group of 12 high-school students reported their weekly study time and grade point average (GPA).

For each student, two numerical variables were recorded:

$$x = \text{weekly study time (hours)}, \qquad y = \text{GPA}.$$

Thus, each student gives one paired observation (x_i, y_i).

```
hours = [6, 14, 3, 22, 9, 11, 12, 5, 18, 24, 15, 17]
gpa   = [2.8, 3.2, 3.1, 3.6, 3.0, 3.3, 3.4, 2.7, 3.1, 3.8, 3.0, 3.9]
```

Formula required

For paired observations

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n),$$

the sample correlation coefficient is

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)s_x s_y}, \qquad -1 \le r \le 1.$$

Here,

$$\bar{x},\ \bar{y}$$

are the sample means, and

$$s_x,\ s_y$$

are the sample standard deviations of the two variables.

The correlation coefficient r describes the **direction and strength of a linear relationship** between two numerical variables.

In general:

- $r > 0$ indicates a positive linear association: larger values of x tend to occur with larger values of y.
- $r < 0$ indicates a negative linear association: larger values of x tend to occur with smaller values of y.
- The magnitude $|r|$ describes the strength of the linear association.
- Values of $|r|$ close to 1 indicate a strong linear association, whereas values close to 0 indicate a weak linear association.

The correlation coefficient should always be interpreted together with a scatter plot.

5A – Study hours and GPA

In this dataset,

$$x = \text{weekly study hours}, \qquad y = \text{GPA}.$$

```
hours = [6, 14, 3, 22, 9, 11, 12, 5, 18, 24, 15, 17]
gpa   = [2.8, 3.2, 3.1, 3.6, 3.0, 3.3, 3.4, 2.7, 3.1, 3.8, 3.0, 3.9]
```

(a) Create a scatter plot with weekly study hours on the horizontal axis and GPA on the vertical axis.

(b) Examine the scatter plot before calculating the correlation. Describe the relationship you observe:

- Does the association appear positive or negative?
- Does the pattern appear approximately linear?
- Are there any observations that appear unusual?

(c) Calculate the sample correlation coefficient r.

(d) Interpret the value of r:

- What does the sign of r tell you?
- What does the magnitude $|r|$ tell you?

(e) Explain in words what the scatter plot and the value of r together tell you about weekly study hours and GPA **in this sample of 12 students**.

(f) Correlation describes an association between variables, but association alone does not establish causation.

Can the correlation coefficient in this sample, by itself, show that studying more *causes* a higher GPA?

Briefly explain why additional information about the study design and other possible factors would be needed to make a causal claim.

5B – Exploring the correlation coefficient

In Part A, the observations do not lie exactly on a straight line. We now use a simple mathematical example to investigate what happens to the correlation coefficient when x and y have an exact linear relationship.

Use the following values of x:

$$x = 1,\ 2,\ 3,\ 4,\ 5,\ 6.$$

First define y by

$$y = 12 - 2x.$$

(a) Calculate the six corresponding y-values and write down the paired observations (x_i, y_i).

(b) Make a scatter plot of the paired observations.

(c) Calculate the correlation coefficient r.

(d) Describe the pattern of the points in the scatter plot. What happens to y as x increases?

(e) What does the **magnitude** $|r|$ tell you about the strength of the linear relationship?

Python

You may use `numpy` and `matplotlib.pyplot` to check your calculations and create the scatter plots.

For example, a scatter plot can be created using

```
import matplotlib.pyplot as plt

plt.scatter(hours, gpa)
plt.xlabel("Weekly study time (hours)")
plt.ylabel("GPA")
plt.title("Study Time and GPA")
plt.grid(True)
plt.show()
```

The correlation coefficient can be calculated using

```
import numpy as np

r = np.corrcoef(hours, gpa)[0, 1]

print("Correlation =", r)
```

Do not report only the Python output. Interpret the correlation coefficient together with the scatter plot.